

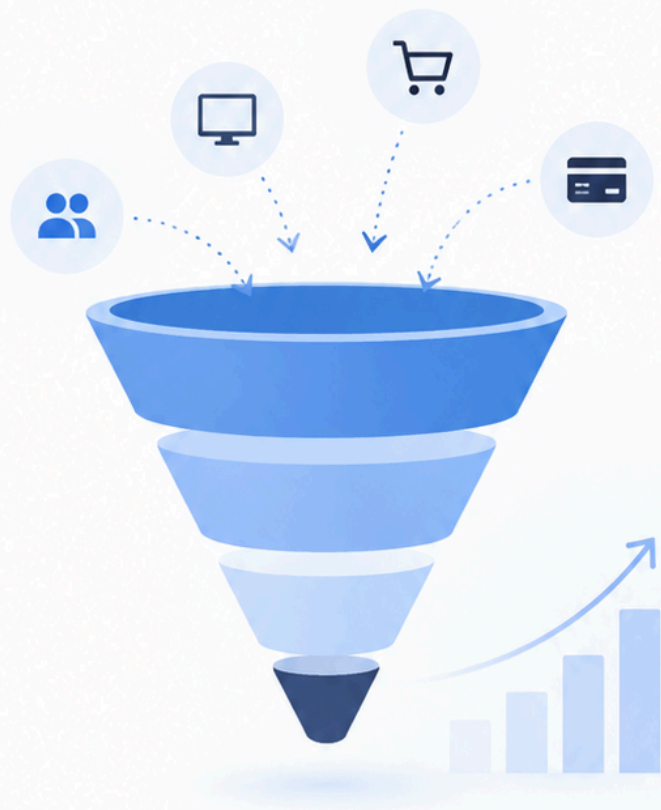
PROJECT REPORT

E-Commerce Funnel Analysis

Data Science and Analytics



DOMAIN
E-Commerce



Prepared by
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2. Project Overview

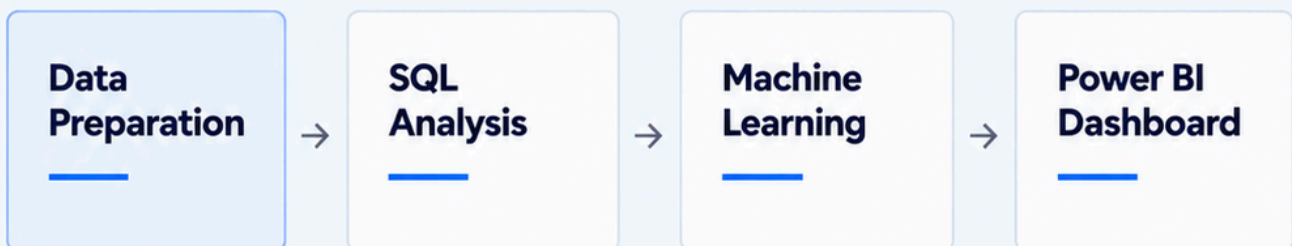
About the Project

This project analyzes the customer journey of an e-commerce website from the first visit to the final purchase. It focuses on understanding funnel drop-offs, conversion performance and revenue efficiency across different channels, campaigns and customer segments.

Project Objectives

- Analyze the conversion funnel from website visit to purchase.
- Identify key drop-off points and conversion rates.
- Evaluate performance across channels, campaigns and customer segments.
- Analyze revenue, discount impact and revenue leakage.
- Build a machine learning model to predict purchase likelihood.
- Create a Power BI dashboard for business insights and decision-making.

Project Workflow



3. Business Problem

Business Problem

E-commerce businesses receive a large number of website sessions, but it is not clearly understood where customers drop off in the purchase journey, which channels and campaigns perform better, and where revenue is being lost.

This makes it difficult to identify key improvement areas and optimize the conversion funnel effectively.

Project Goal

To analyze the e-commerce conversion funnel, identify drop-off points, evaluate channel and campaign performance, analyze revenue efficiency, and develop a machine learning model to predict purchase likelihood.

4. Dataset Overview

Dataset

The dataset contains e-commerce website session data with customer, marketing, funnel and revenue information.

DATASET SIZE

120,000 sessions

Key Columns

Column	Description
user_id	Unique customer identifier
session_id	Unique session identifier
date	Session date
channel	Acquisition channel (e.g., Paid Ads, Email, Organic, Social)
campaign_type	Campaign type
user_type	New or Returning user
device	Device used (Mobile or Desktop)
region	Customer region
visited_website	Website visit flag (Yes/No)
viewed_product	Product view flag (Yes/No)
added_to_cart	Add to cart flag (Yes/No)
checkout_started	Checkout started flag (Yes/No)
purchase_completed	Purchase completed flag (Yes/No)
discount_applied	Discount applied flag (Yes/No)
order_value	Original order value
revenue	Final revenue after discount

5. Data Cleaning & Preparation

1. Load Dataset

```
[1] import pandas as pd
df = pd.read_csv(r'../data/raw.csv')

[2] df.shape
(120000, 17)
```

2. Check Missing Values

```
[3] df.isnull().sum()

... user_id      0
     session_id  0
     date        0
     month       0
     channel     0
     campaign_type 0
     ...
```

3. Check Funnel Consistency

```
[7] df[(df['viewed_product']=='No') & (df['added_to_cart']=='Yes')]
     user_id session_id date month channel campaign_type device user_type ...

[8] df[(df['added_to_cart']=='No') & (df['checkout_started']=='Yes')]
     user_id session_id date month channel campaign_type device user_type ...

[9] df[(df['checkout_started']=='No') & (df['purchase_completed']=='Yes')]
     user_id session_id date month channel campaign_type device user_type ...
```

4. Revenue Check

```
[10] df[df['purchase_completed']=='No']['revenue'].sum()
0.0
```

5. Convert Date Format

```
[11] df['date'] = pd.to_datetime(df['date'],format='%m/%d/%Y')
```

Cleaning Steps

- Loaded the dataset and checked its shape.
- Checked for missing values in all columns.
- Validated funnel logic to ensure no invalid stage progression.
- Verified that revenue is only present for completed purchases.
- Converted the date column to datetime format.

Dataset After Cleaning

The dataset is clean and ready for analysis.

Total Sessions	120,000
Total Columns	17
Missing Values	0
Invalid Funnel Rows	0
Revenue (No Purchase)	0
Date Format	Converted to datetime

Result

The dataset is clean with no missing values, consistent funnel behaviour and valid revenue information, and is ready for further SQL analysis, machine learning and dashboard development.

6. Machine Learning Model

A machine learning classification model was developed to predict whether a website session will result in a purchase. Traditional analysis in SQL and Power BI shows past trends, but it cannot predict which ongoing sessions are likely to convert.

The model uses session behaviour and customer attributes to estimate the purchase probability. This helps in identifying high-potential users, improving marketing targeting and optimizing conversion strategies.

Multiple algorithms were trained and compared, including Logistic Regression, Random Forest and XGBoost. Categorical variables were converted to numerical format using one-hot encoding and the data was split into training and testing sets for model evaluation.

```
[51] X = df[['channel', 'campaign_type', 'device', 'user_type', 'region',  
        'visited_website', 'viewed_product', 'added_to_cart', 'checkout_started']]  
y = df['purchase_completed']
```

```
[52] X = pd.get_dummies(X, drop_first=True, dtype=int)  
y = pd.get_dummies(y, drop_first=True, dtype=int)
```

```
[53] df['added_to_cart'] = df['added_to_cart'].map({'No':0, 'Yes':1})  
df['checkout_started'] = df['checkout_started'].map({'No':0, 'Yes':1})  
df['visited_website'] = df['visited_website'].map({'No':0, 'Yes':1})  
df['viewed_product'] = df['viewed_product'].map({'No':0, 'Yes':1})  
df['purchase_completed'] = df['purchase_completed'].map({'No':0, 'Yes':1})  
  
bool_cols = df.select_dtypes(include='bool').columns  
df[bool_cols] = df[bool_cols].astype(int)
```

```
[54] from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
[55] from sklearn.linear_model import LogisticRegression  
  
model = LogisticRegression()  
model.fit(X_train, y_train)  
y_pred_lr = model.predict(X_test)
```

```
[90] from sklearn.ensemble import RandomForestClassifier  
  
rf_model = RandomForestClassifier(  
    n_estimators=100,  
    max_depth=8,  
    min_samples_split=5,  
    min_samples_leaf=2,  
    random_state=42,  
    class_weight='balanced'  
)  
  
rf_model.fit(X_train, y_train)  
y_prob_rf = rf_model.predict_proba(X_test)[:, 1]  
threshold_rf = 0.92  
y_pred_rf_threshold = (y_prob_rf >= threshold_rf).astype(int)
```

```
[60] from xgboost import XGBClassifier  
  
xgb_model = XGBClassifier(  
    n_estimators=100,  
    max_depth=3,  
    learning_rate=0.08,  
    random_state=42,  
    eval_metric='logloss'  
)  
  
xgb_model.fit(X_train, y_train)  
y_pred_xgb = xgb_model.predict(X_test)  
y_prob_xgb = xgb_model.predict_proba(X_test)[:, 1]
```


7. Model Results & Comparison

Three classification models were trained and evaluated to predict whether a website session will result in a purchase.

The models were compared using standard evaluation metrics to find the best performing model.

Logistic Regression

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train,y_train)
y_pred_lr = model.predict(X_test)
```

Accuracy	0.9336
Precision	0.4998
Recall	0.6319
F1 Score	0.5581
ROC AUC	0.7935

Confusion Matrix

```
confusion matrix
[[21401  1007]
 [   586  1006]]
```

Random Forest

```
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    n_estimators=100,
    max_depth=8,
    min_samples_split=5,
    min_samples_leaf=2,
    random_state=42,
    class_weight='balanced'
)

rf_model.fit(X_train,y_train)
```

Accuracy	0.9343
Precision	0.5028
Recall	0.8411
F1 Score	0.6294
ROC AUC	0.9640

Confusion Matrix

```
confusion matrix
[[21084  1324]
 [   253  1339]]
```

XGBoost

```
from xgboost import XGBClassifier

xgb_model = XGBClassifier(
    n_estimators=100,
    max_depth=3,
    learning_rate=0.08,
    random_state=42,
    eval_metric='logloss'
)

xgb_model.fit(X_train,y_train)
```

Accuracy	0.9353
Precision	0.5090
Recall	0.6778
F1 Score	0.5814
ROC AUC	0.8157

Confusion Matrix

```
confusion matrix
[[21367  1041]
 [   513  1079]]
```

Why Random Forest?

Random Forest was selected as the final model because it achieved the highest recall and ROC AUC among all models. Recall is important in this case because the goal is to identify as many potential purchase sessions as possible. It also provides a good balance between precision and recall compared to Logistic Regression and XGBoost.

Final Model

Model	Random Forest
Key Metrics	Accuracy: 0.9343 Recall: 0.8411 ROC AUC: 0.9640
Reason	Highest recall and ROC AUC, better balance between precision and recall

8. Streamlit Web Application

The trained Random Forest model was deployed using a Streamlit web application. The application allows users to input session and customer details and returns the purchase probability along with the prediction result.

E-commerce Purchase Prediction

Predict the probability of a customer session resulting in a purchase.

Machine Learning | Random Forest

Customer Session Details

Channel

Paid Ads

Campaign Type

New Launch

Device

Desktop

User Type

New User

Visited Website

Yes

Viewed Product

Yes

Added to Cart

Yes

Checkout Started

Yes

Region

Non-Metro

Predict Purchase

Purchase Predicted

Purchase Predicted

PURCHASE PROBABILITY

94.03%

Decision Threshold: 92% Purchase is predicted when the model probability is at least 92%.

Prediction Summary

PREDICTION	PROBABILITY	DECISION THRESHOLD
Purchase Predicted	94.03%	92%

```
import streamlit as st
import joblib

model = joblib.load("../models/rf_purchase_model.pkl")

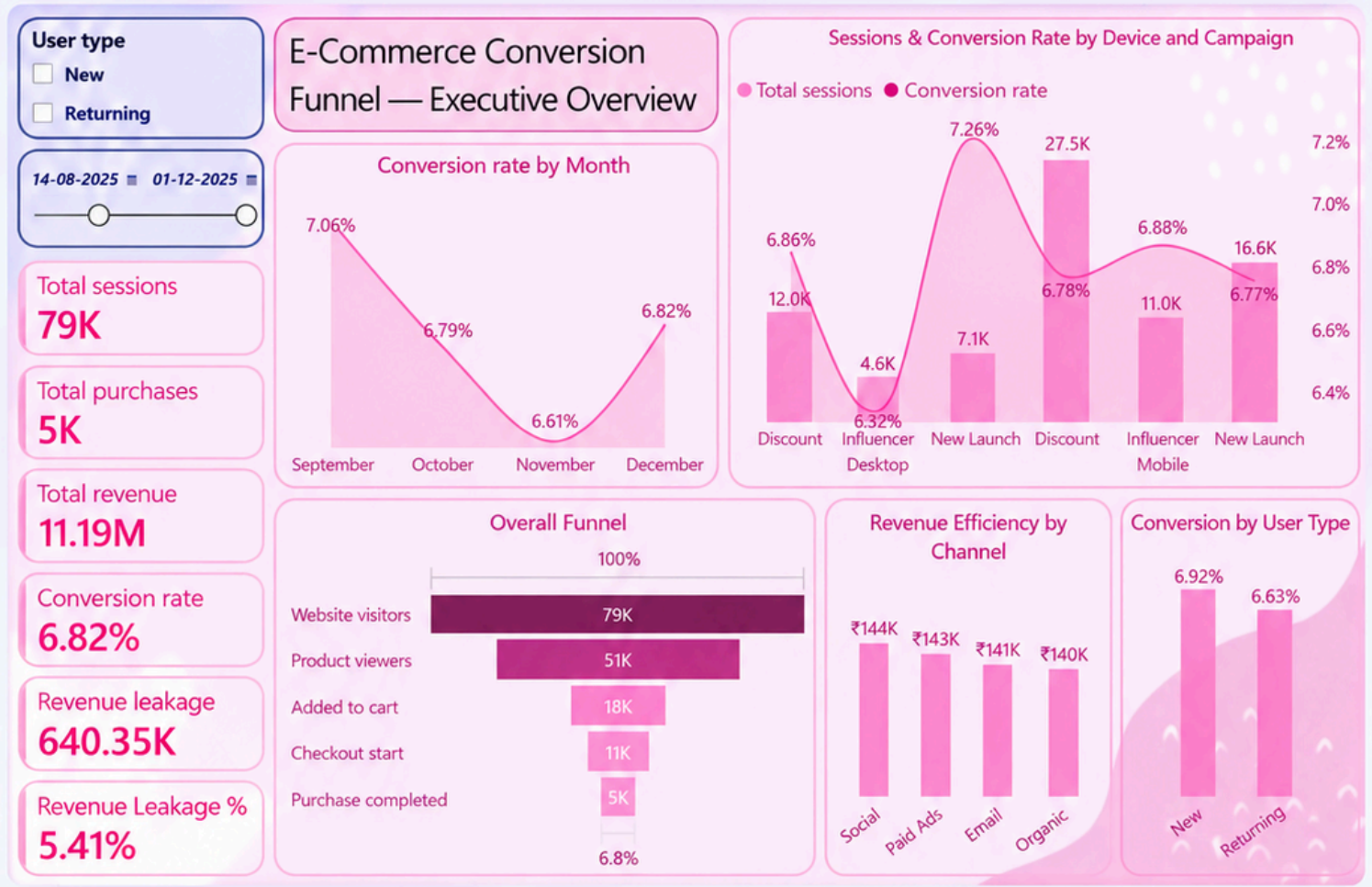
st.title("E-commerce Purchase Prediction")
st.write("Predict the probability of a customer session
resulting in a purchase.")
st.subheader("Customer Session Details")
```

```
# Get user input
channel = st.selectbox("Channel", ["Paid Ads", "Organic", "Email"])
campaign = st.selectbox("Campaign Type", ["New Launch", "Discount"])
device = st.selectbox("Device", ["Desktop", "Mobile"])
user_type = st.selectbox("User Type", ["New User", "Returning User"])
region = st.selectbox("Region", ["Metro", "Non-Metro"])

# Predict
input_data = pd.DataFrame([user_input_dict])
prob = model.predict_proba(input_data)[0][1]
pred = "Purchase" if prob >= 0.92 else "No Purchase"
st.write(f"Purchase Probability: {prob:.2%}")
```


9. Power BI Dashboard 1

An interactive dashboard was created to analyze e-commerce performance and customer behavior. It provides key metrics and visual insights with filters, helping to monitor trends and identify opportunities for improving conversion and revenue.



What the Dashboard Shows?

The dashboard visualizes key e-commerce metrics such as total sessions, purchases, revenue, conversion rate and revenue leakage. It includes interactive filters for user type and date range, and provides detailed breakdowns by month, device, campaign and channel. The funnel view shows the drop-off at each stage, helping to identify where users are leaving. These insights support better decision-making for marketing and optimization.

Business Impact

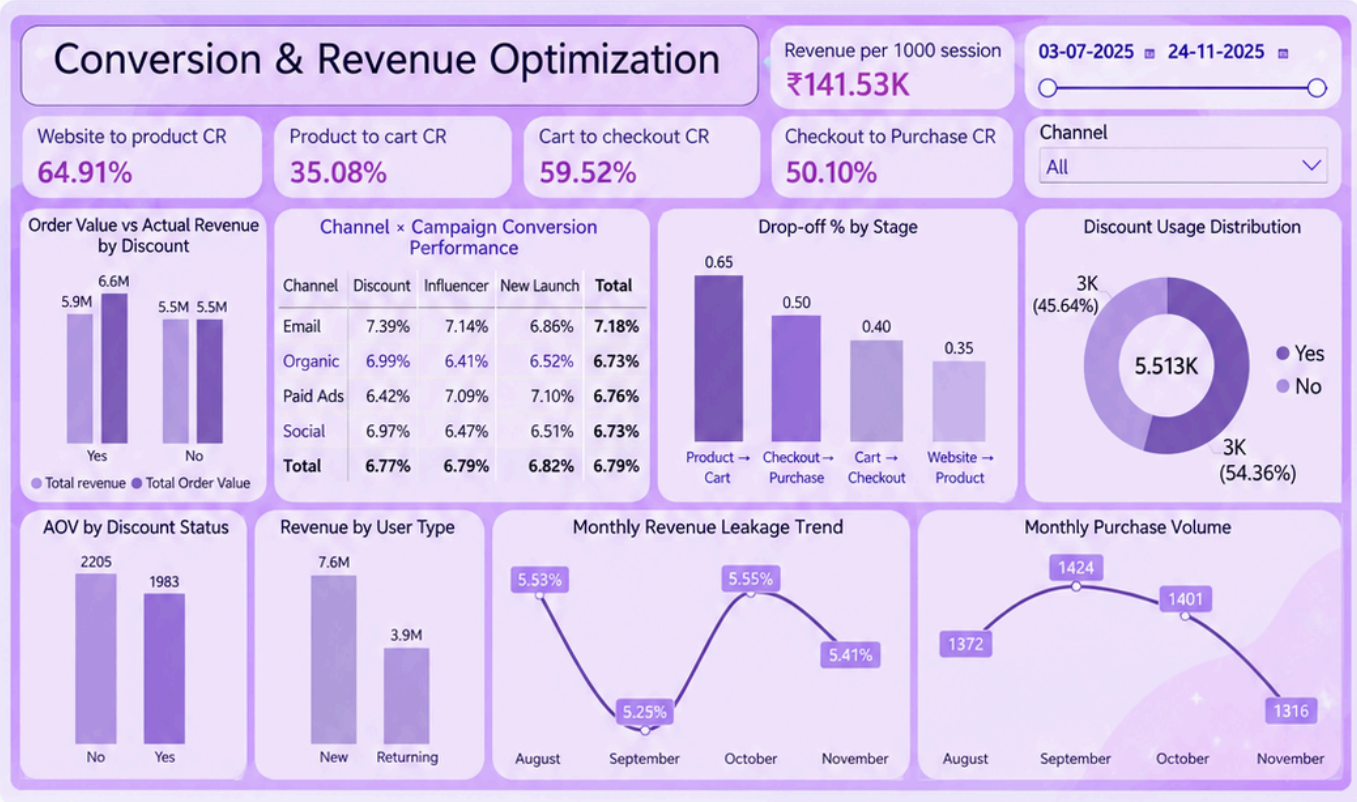
The dashboard helps track campaign performance, identify high-performing segments and detect drop-off points in the funnel. This enables data-driven decisions to improve conversion rate, reduce revenue leakage and increase overall revenue.

Key Insights

- Conversion rate varies by month, with a peak of 7.26%.
- New users convert slightly better (6.92%) than returning users (6.63%).
- Discount campaigns and mobile devices drive higher traffic.
- Major drop-off occurs between product view and add to cart stage.

10. Power BI Dashboard 2

A second interactive dashboard was built to analyze conversion performance and identify opportunities for revenue optimization. It provides detailed stage-wise conversion rates, revenue per session and campaign performance with dynamic filters.



Dashboard Overview

This dashboard focuses on conversion and revenue optimization. It tracks key stage-wise conversion rates from website visit to purchase, compares revenue performance across campaigns and discount usage, and highlights areas where users drop off. Filters for date range and channel allow dynamic analysis of different segments.

Business Impact

The dashboard helps in identifying high-performing campaigns, understanding the impact of discounts, and reducing revenue leakage at critical stages. These insights support data-driven decisions to optimize marketing strategies and improve overall conversion and revenue.

Key Insights

- Website to product conversion rate is 64.91%.
- Cart to checkout conversion rate is 59.52%.
- Revenue per 1000 sessions is ₹141.53K.
- Monthly revenue leakage remains between 5.25% – 5.55%.
- Discount campaigns and new users drive higher purchase volume.

11. Key Business Insights

Overall Performance

The analysis shows a total of 79K sessions with 5K purchases, generating a revenue of 11.19M. The overall conversion rate is 6.82% and revenue leakage is 640.35K, which indicates a good opportunity to improve conversion and recover potential revenue.

Monthly Trend

Conversion rate varies by month, with the highest conversion rate of 7.06% in September and the lowest of 6.61% in November. Revenue leakage percentage ranges between 5.25% and 5.55%, showing some fluctuation across months.

Channel Performance

Channels like Discount, Influencer and New Launch drive higher traffic and conversions. Based on the analysis, campaign performance differs across channels, with some channels generating higher order value and better conversion rates.

Device and User Behavior

Desktop and mobile devices contribute a significant number of sessions. Mobile devices show strong engagement and conversions, especially in campaigns like Influencer and New Launch. New users have a slightly higher conversion rate (6.92%) compared to returning users (6.63%).

Funnel Analysis

The funnel analysis shows the drop-off at each stage. Website to product conversion rate is 64.91%, product to cart is 35.08%, cart to checkout is 59.52%, and checkout to purchase is 50.10%. The major drop-off occurs between product view and add to cart stage, which indicates a need to improve product engagement.

Revenue Optimization

Average revenue per 1000 sessions is 141.53K. Discount usage contributes a significant portion of orders, with around 45.64% of purchases using discounts. Analyzing discount strategies and campaign performance can help increase overall revenue and reduce leakage.

Business Impact

These insights help in identifying high-performing segments, improving campaign strategies, optimizing the user journey and reducing revenue leakage, which can lead to higher conversion rates and increased revenue for the business.

12. Business Recommendations

Based on the analysis from the dashboards and machine learning model, the following recommendations can help improve conversion rate and overall revenue.

Improve Product Engagement

A significant drop-off is observed between product view and add to cart stage. This indicates that users are viewing products but not adding them to the cart. Improving product images, descriptions, reviews and personalized recommendations can help increase engagement and add to cart rate.

Optimize Marketing Campaigns

Channels like Discount, Influencer and New Launch show higher traffic and conversions. The business can focus more on these high-performing campaigns and allocate more budget to them. Underperforming channels can be reviewed and optimized or reduced.

Target New Users

New users have a slightly higher conversion rate (6.92%) compared to returning users (6.63%). This suggests that acquisition campaigns are effective. The business should continue to attract new users while also running targeted campaigns to re-engage returning users.

Leverage Discount Strategies

Around 45.64% of purchases are made using discounts. Well-planned discount strategies can further increase purchase volume, but discounts should be optimized to maintain profitability.

Focus on Mobile Users

Mobile devices contribute a significant number of sessions and show strong engagement. Ensuring a smooth and optimized mobile shopping experience can help improve conversion rates.

Reduce Revenue Leakage

Revenue leakage of 5.41% indicates potential lost revenue. Analyzing drop-off points in the funnel, improving the checkout process and addressing cart abandonment can help recover this revenue.

Use ML Model for Personalized Targeting

The trained Random Forest model can be used to identify high-intent users and provide personalized offers, product recommendations and targeted marketing, which can improve conversion rate and increase overall revenue.

13. Project Limitations & Conclusion

Project Limitations

The analysis is based on a sample e-commerce dataset, which may not fully represent real business data. Some important factors such as product categories, customer demographics and external market trends are limited in the dataset.

The machine learning model is trained on historical data, so its predictions may not always be accurate for new or unseen patterns. The dashboard also focuses on key metrics and does not include advanced customer segmentation or real-time data.

Conclusion

This project provides valuable insights into e-commerce performance, customer behavior and conversion patterns. The data analysis and interactive dashboards help in identifying key trends, high-performing channels and areas of improvement. The machine learning model further supports the analysis by predicting the likelihood of purchase, which can be used for targeted marketing and personalized recommendations.

Overall, the project demonstrates how data analytics and machine learning can be combined to support data-driven decision making, improve conversion rates and increase revenue for an e-commerce business.

